

Multiple Choice (1 point each)

Question 1

Dynamic programming is useful for preventing duplicate function calls.

- ☐ True
 - ☐ False
-

Question 2

NP-complete problems are reducible to each other.

- ☐ True
 - ☐ False
-

Question 3

Problems in class NP can be solved in polynomial time.

- ☐ True
 - ☐ False
-

Question 4

In asymptotic ordering, exponential time is less than polynomial time.

- ☐ True
 - ☐ False
-

Question 5

Hamiltonian Circuit and Hamiltonian Path are reducible to each other.

- ☐ True
 - ☐ False
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Question 6

The 3SAT problem is NP-complete.

- ☐ True
 - ☐ False
-

Question 7

The Cook–Levin theorem provides the foundation for NP-complete problems.

- ☐ True
 - ☐ False
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Question 8

The k-Clique problem is reducible to the 3SAT problem.

- ☐ True
 - ☐ False
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Question 9

The matrix chain multiplication (parenthesization) problem can be solved using dynamic programming.

- ☐ True
 - ☐ False
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Question 10

The class NP-complete is contained in NP and not in NP-hard.

- ☐ True
- ☐ False